

Geometry-Sensitive Stabilization via Structured Finite-Rank Compression in Non-Normal Operators

An Exploratory Computational Study

Abstract

Transient amplification is a characteristic feature of non-normal operators. This exploratory computational study compares a structured finite-rank projection with rank-matched random projections and synthetic low-commutator controls. The experiments suggest that the structured projection consistently reduces transient amplification and related stability measures. A reduced 2×2 Jordan model reproduces the same qualitative behaviour. The work is limited to finite-dimensional computational models and proposes a candidate geometric mechanism for future investigation.

Introduction

Many non-normal operators exhibit large transient responses despite asymptotic stability. This work investigates whether carefully designed low-rank projections can suppress this behaviour more effectively than generic projections.

Method

Three projection families are compared: a structured projection, Haar-random rank-matched projections, and synthetic low-commutator projections. The study evaluates the Kreiss constant, Henrici departure from normality, pseudospectral area, commutator defect (δ), and geometric overlap (η).

Results

Across the numerical experiments, the structured projection consistently produced lower transient amplification than the comparison methods. Knockout experiments and a reduced 2×2 model support the proposed geometric interpretation.

Discussion

The observations suggest that stabilization depends on both a small commutator defect and a small geometric overlap with dominant transient amplification directions. These findings are computational observations rather than mathematical proofs.

Limitations

This work is exploratory. No claims are made regarding the Riemann zeta function, the Riemann Hypothesis, or infinite-dimensional operator theory. Independent

verification and further mathematical analysis are required.

Conclusion

Structured finite-rank compression appears to reduce transient amplification more effectively than matched random or synthetic projections in the computational models studied. Further theoretical and experimental work is needed to determine the generality of this mechanism.